Unusual Does Not Mean Impossible

AP Statistics · Unit 2 · Topic 2.10 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Using Probability and Simulation: Determine parameters for probability distributions.
- **Skill 4.B** — Statistical Argumentation: Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.C** — Calculate parameters for a binomial distribution. [Skill 3.B]
- **EK UNC-3.C.1** — If a random variable is binomial, its mean, μ_X, is np and its standard deviation, σ_X, is √(np(1 - p)).
- **LO UNC-3.D** — Interpret probabilities and parameters for a binomial distribution. [Skill 4.B]
- **EK UNC-3.D.1** — Probabilities and parameters for a binomial distribution should be interpreted using appropriate units and within the context of a specific population or situation.

Essential question: How do binomial parameters help distinguish an unusual count from an impossible one?

DOK-3 skill: 4.B · **FRQ pattern:** `binomial-mean-sd-unusual-vs-impossible`

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Compares 58 with mean 70 and SD about 4.6 in count units
- **reasoning-2** (required) — Uses the positive 0.0072 chance to distinguish unusual from impossible
- **reasoning-3** (required) — Supports the analyst and explains the coach's false guaranteed cutoff or proof claim

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Using 0.70 as the mean count instead of 70
- Omitting the square root in the standard deviation
- Treating an unusual outcome as impossible or as proof that the model changed

[STAR] THE PROBLEM — annotated

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: "Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies."

Analyst: "A result can be unusual without being impossible; compare 58 with the model's mean and standard deviation."

The binomial model gives a chance of about 0.0072 for 58 or fewer makes; use this provided value.

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE — one sentence before you work the parts

Whose statement is more defensible, the coach's or the analyst's? Name one quantity you would check.
Key: Not graded. Listen for whether students use impossible to mean unlikely; the calculation should sharpen that language.

[BOOK] CENTER, SPREAD, AND UNUSUAL COUNTS (keep on the page)

For a binomial count, $\mu_X = np$ and $\sigma_X = \sqrt{np(1-p)}$, both in count units. The mean is an average over many comparable sets of trials. The standard deviation describes a typical distance from that mean. A result far from the mean may be unusual, but a small positive probability does not mean impossible or prove that the model changed.

DOK 1 **(a)** State n and p for this model.

Key: $n = 100$ attempts and $p = 0.70$ for each attempt.

DOK 2 **(b)** Calculate the binomial mean and standard deviation. Interpret the mean for many sets of 100 shots.

Key: $\mu_X = 100(0.70) = 70$ makes and $\sigma_X = \sqrt{100(0.70)(0.30)} = \sqrt{21} \approx 4.6$ makes. Across many comparable sets of 100 attempts, the model predicts an average of 70 makes.

DOK 3 **(c)** Whose statement is supported? Compare 58 with the mean and standard deviation, and use the given probability to distinguish unusual from impossible. Explain what the coach's claim could make a reader wrongly believe. Write 3–4 sentences.

Key: The analyst is supported: 58 is 12 makes below the mean of 70, more than twice the typical distance of about 4.6 makes. The given chance of 58 or fewer is about 0.0072, so such a low result is unusual but has positive probability. The coach turns a typical performance level into a guaranteed lower bound. That could make a reader think one unusual set proves the career rate changed, even though chance can produce it.